

ResuMetrics

Project Summary & Architecture Overview

1. Overview & Purpose

ResuMetrics is an AI-powered hybrid resume screening and candidate matching web application. It is designed for HR teams, hiring managers, and recruiters who need to process large volumes of resumes quickly. The primary goal is to analyze candidate resumes (PDF or DOCX) against a job description or skill list and generate a ranked leaderboard based on requirements matching.

2. Core Functionality & User Workflow

- **Input:** Upload multiple candidate resumes via a drag-and-drop interface and paste a job description.
- **Customization:** Tweak the importance of different matching algorithms using weight sliders.
- **Processing:** Extracts text directly from files in memory, cleans the text, and runs it through multiple AI scoring algorithms in parallel.
- **Results:** A dashboard displays interactive cards for each candidate featuring score gauges, matched/missing skill badges, contact info, and fraud alerts.
- **Export:** The final leaderboard can be exported as a professional PDF report or a CSV spreadsheet.

3. The 'AI Brain' (Scoring Algorithms)

The hybrid scoring engine blends three independent algorithms to ensure fair and accurate candidate ranking:

- **TF-IDF Keyword Matching (40%):** Finds overlapping technical terms and keywords, emphasizing rare, distinct terms over common filler words.
- **Exact Skill Matching (35%):** A fast algorithm that checks for the binary presence of specific required skills.
- **Dense Semantic Matching (25%):** Utilizes SentenceTransformer (an AI neural network) to understand the meaning of the text, matching semantically similar concepts.

4. Advanced NLP & Smart Features

- **Timeline Fraud Detection:** Compares explicit years of experience claimed against actual dates in job history, raising an alert on discrepancies.
- **Named Entity Recognition (NER):** Automatically categorizes degrees, job roles, professional certifications, and spoken languages.
- **Behavioral & Metric Extraction:** Identifies soft skills, impact-driven action verbs, and quantifiable metrics (e.g., '\$10M', '50%').

- **RAG AI Chatbot:** Integrated chatbot allows recruiters to ask semantic questions over all uploaded candidate resumes.
- **Kanban ATS Pipeline:** Interactive drag-and-drop board to track candidates through Applied, Shortlisted, Interviewing, and Rejected stages.
- **Automated Shortlisting:** Integrated SMTP mailer to bulk-email selected candidates directly from the dashboard.

5. Architecture & Tech Stack

- **Backend (Python):** Flask for web server, SQLite for data, PyPDF/python-docx for parsing, and ReportLab/Pandas for exporting.
- **AI Engines:** scikit-learn, sentence-transformers, PyTorch, and NumPy handle the advanced scoring and matching.
- **Frontend:** HTML5, CSS3, Vanilla JavaScript, and SVG for animations, featuring a responsive, modern UI.

6. Performance Optimizations

- **In-Memory Processing:** Eliminates disk I/O by reading file bytes directly.
- **Concurrency:** Uses ThreadPoolExecutor to parse multiple resumes simultaneously.
- **AI Model Pre-warming:** The SentenceTransformer model loads in a background thread at startup for instant readiness.
- **Caching:** Uses SHA-256 to cache semantic vectors for resumes, making repeated analyses instant.
- **Single-Pass Regex:** Consolidates multiple regex replacement passes into a single pass for faster text normalization.